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DAA LAB

LAB EXPERIMENTS

EXPERIMENT – 3

SELECTION SORT ON RANDOM, REVERSE AND SORTED ARRAYS

AIM:

To implement the Selection Sort algorithm in Python and sort random, reverse-ordered, and already sorted arrays in ascending order.

ALGORITHM:

1. Start the program.
2. Initialize three arrays: random, reverse-ordered, and sorted.
3. Find the minimum element in the unsorted portion of the array.
4. Swap the minimum element with the first element of the unsorted portion.
5. Repeat the process until the entire array is sorted.
6. Display the sorted arrays.
7. Stop the program.

PROCEDURE:

1. Open a Python IDE or editor.
2. Create three arrays with different initial orders.
3. Define a function to perform Selection Sort.
4. Compare elements and identify the minimum element in each pass.
5. Swap the minimum element into its correct position.
6. Execute the program and observe the output.

PYTHON CODE:

```python
def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_index = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        arr[i], arr[min_index] = arr[min_index], arr[i]
    return arr
random_arr = [64, 25, 12, 22, 11]
reverse_arr = [5, 4, 3, 2, 1]
sorted_arr = [1, 2, 3, 4, 5]
print("Random Array:", selection_sort(random_arr))
print("Reverse Array:", selection_sort(reverse_arr))
print("Sorted Array:", selection_sort(sorted_arr))
```

SAMPLE OUTPUT:

```
Output

Random Array: [11, 12, 22, 25, 64]
Reverse Array: [1, 2, 3, 4, 5]
Sorted Array: [1, 2, 3, 4, 5]
```

RESULT:

The Selection Sort algorithm was successfully implemented in Python. The random, reverse-ordered, and already sorted arrays were sorted in ascending order.